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Ultrastrong dissimilar glass-ceramic to Fe–36Ni alloy joints using ultrafast laser micro-welding

Tao Zhang ^a, Jin Yang ^a  , Meng Yang ^a, Ming Wu ^a, Rui Zhou ^a, Jianian Tian ^a, Yixuan Zhao ^a, Caiwang Tan ^b, Xiaoguo Song ^b [Show more](#) ▼

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Highlights

- Ablation threshold regulation was proposed to control energy deposition across dissimilar materials for ultrastrong micro-welds.
- The transition from thermal-conduction-dominated to nonlinear absorption-synergized bonding regimes was realized.
- The mechanism of the influence of laser parameters on the whisker evolution process was revealed.
- Phase transformation and interlocking whisker enabled multi-scale crack deflection and bridging for interfacial strengthening.

Abstract

Ultrastrong Fe—36Ni alloy/glass-ceramic joints were successfully fabricated using ultrafast laser micro-welding technology with the aid of material ablation threshold regulation strategy for the first time. High-aspect-ratio LiAlSi₂O₆ whiskers were identified at the interface. The microstructural evolution and strengthen effects to the ultrafast laser micro-welds was systematically investigated. As the laser energy density increased from 1.16J/cm² to 5.41J/cm², the micro-welds morphology changed from metal-dominated thermal conduction behaviors (splashing, wetting, and adhesion) to nonlinear absorption-dominated mechanical interlocking with volcanic-like features. With the increase of laser energy density, the length and width of LiAlSi₂O₆ whiskers in the micro-welds increased and interlaced each other to form a mechanical interlock structure, resulting in a maximum shear strength of 44MPa. This demonstrates the strengthen effect of the whiskers on the joint strength. The findings hold significant implications for ultrafast laser welding technology between glass-ceramic and metal, providing technical support for the fabrication of high-performance space mirror components.

Graphical abstract

5. Conclusion

CRedit authorship contribution statement

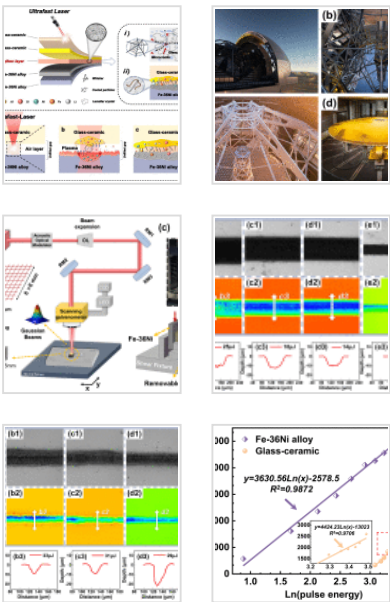
Declaration of competing interest

Acknowledgments

References

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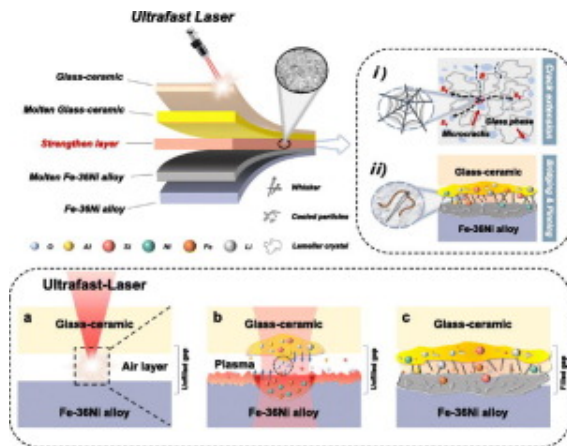


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Keywords

Ultrafast laser micro-welding; Glass-ceramic; Fe-36Ni alloy; Laser-induced ablation threshold; Strengthen mechanism; Evolution mechanism

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1. Introduction

Lithium-aluminum-silicate (LAS) glass-ceramic consists of a glassy phase that is interspersed with tiny crystals. It combines the advantages of glasses and ceramics, offering higher thermal and mechanical stability, as well as ensuring a dense surface structure and excellent optical properties. It has been widely used for micro-electro-mechanical systems (MEMS), microlithography instruments, space remote sensing detectors, ring laser gyroscopes, and other types of composite components with metal as a continuum, which is capable of meeting the requirements of aerospace, microprisms

[Table 3. LIAT regulation strategy for the ...](#)

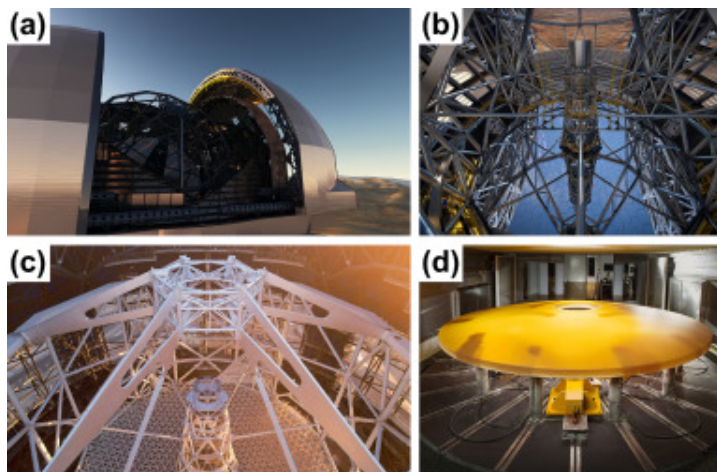
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and other applications [[1], [2], [3], [4], [5], [6]]. As a highly lightweight components, especially with respect to the harsh space environment, LAS glass-ceramic such as Zerodur® has emerged as an attractive option for satellite optical imaging systems due to its extremely low coefficient of linear thermal expansion (CTE), excellent chemical stability, and high reflectivity in specific optical bands, such as Hubble Space Telescope, Spinning Enhanced Visible and Infrared Imager (SEVIRI) [3,7,8]. To achieve reliable components in the above-mentioned applications, the high strength bonding between glass-ceramic to other kinds of materials such as metals is very urgent [[9], [10], [11], [12]]. Fig. 1 shows the macrographs and joint configurations for Fe-36 Ni/glass-ceramic dissimilar components.



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Fig. 1. (a) European extremely large optical/infrared telescope (E-ELT) (ESO/L. Calçada). (b) E-ELT dome and telescope structure (ESO/L. Calçada/ACe Consortium). (c) the inner skeleton of E-ELT (ESO/G. Vecchia). (d) the E-ELT's M2 Mirror Blank (SCHOTT ZERODUR®).

However, the significant differences between metals and glass-ceramics in physical and chemical properties have been identified as the primary constraint, which prevent the application of conventional fusion welding techniques. Although multiple advanced

bonding approaches, including brazing, solid state bonding, ultrasonic welding, anodic bonding, and cementation were systematically investigated [[13], [14], [15], [16], [17], [18], [19]], they were proven inadequate due to limitations: (i) inefficient interfacial reactions, (ii) poor precision, (iii) high residual stress and (iv) the addition of extra interlayer addition, which restrict the further application of the components. For instance, in vacuum brazing of SiO₂ glass-ceramic/TC4 alloy, Liu et al. [17,19] found that the holding time seriously affected the joint strength, that was insufficient holding time induced incomplete interfacial reactions, which failed to bear high strength. Conversely, a long-holding time was easily led to the formation of high residual stresses at the interface, The optimized brazing at 900°C/5min achieved peak shear strength of 27MPa. Lascar et al. [16] achieved glass-ceramic/Al/Invar alloy junctions by solid state diffusion bonding. The results showed that yield residual stresses were produced due to expansion mismatch, which was likely to cause ceramic flaws to propagate after cooling or thermal cycling. Gester et al. [14] utilized ultrasonic torsion welding of CERAN® glass-ceramics and Kovar alloys with a thin aluminum interlayer, it was found that the presence of residual stresses damaged the aluminum residues remaining on the CERAN® surface, which reached a poor characteristic shearing force for joints. Anodic bonding of stainless steel to Li₂O-ZnO-SiO₂ glass-ceramic were performed successfully by Xiong et al. [18]. Nevertheless, this method imposed surface roughness requirements, which elevated the overall production expenses. The resultant bond strength was only 1.82MPa. Furthermore, the drawback of conventional techniques employing polymer-based adhesives lies in their necessitating curing period, not to mention the effects of aging, especially in the case of fluctuating thermal conditions, which weaken the tensile strength of the bond [13,15]. In conclusion, the afore-mentioned methods are insufficient to obtain high-performance glass-ceramic to metal dissimilar components for advanced optoelectronic devices. Thus, an alternative joining technology with high strength and reliability is highly expected.

In recent years, great interests have been attracted in ultrafast laser because of its extremely high peak power (greater than several tens of TW/cm²) and extremely small temporal scales (10⁻¹² to 10⁻¹⁵s) [20,21]. Ultrafast laser is very promising in the joining of

transparent materials taking advantage of the unique nonlinear absorption mechanism, enables direct energy deposition and local modification [[22], [23], [24], [25]]. The resulting volume of molten glass generated, although small, is sufficient to fill the micron-scale gap. This is due to the fact that the high-temperature plasma is restrained and adheres to the underside of the glass, which begins to melt occurred by heat conduction, while the melt flow is driven by thermal expansion, and eventually solidifies to form the joint. In fact, ultrafast laser welding technology has also been used for the joining of transparent materials with metals since 2008 [26]. Ozeki et al. [26] were the first to successfully join glass and copper by femtosecond (fs) pulsed laser without any voids or cracks at the interface, and the joint strength was up to 16MPa. In contrast to nanosecond (ns) pulses, femtosecond lasers enabled precise control of the weld area through thermal suppression and a high degree of spatial selectivity [27,28]. Extensive studies were pursued by scholars aiming to achieve high-strength joints. For instance, Zhang et al. [24] used femtosecond pulse laser to directly weld metal and glass by varying the pulse power and scanning speed, and the bonding strength characterized was as high as 2.34MPa. Wang et al. [29] used to join 304 stainless steels and glasses by femtosecond laser. The influence of laser energy inputs, defocusing distance, and welding velocity on the weld quality were illustrated. The joint strength at maximum was achieved 8.79MPa. Yang et al. [30] investigated the relationship between surface roughness and the mechanical behavior of sapphire/Invar joints using femtosecond laser. The strength of the joints increased then decreased with the lowered Sa . Maximum shear strength of 107.87MPa was obtained at the $Sa \sim 0.131 \mu\text{m}$. Zhang et al. [31] proposed a mechanical v-pin structure at the interface which achieved a tight locking between glass and aluminum—greatly improving the bonding strength to 100.91 MPa. Overall, the afore-mentioned studies collectively contributed to provide ideas for joints reinforcement by basic process parameters or interface regulation in ultrafast laser welding techniques, but all the works lacked of referring the process parameters to ablative properties of the considered materials. In the case of transparent materials, such as ceramics, achieving localized melting within a specific processing window, rather than ablation, is more favorable for achieving effective joining [32]. It is therefore necessary to optimize the application of

ultrafast laser welding technology in a variety of applications by establishing a correlation between laser and material properties.

In the present study, ultrastrong Fe—36Ni alloy/glass-ceramic joints were successfully produced by ultrafast laser micro-welding technology, with the aid of material ablation threshold regulation strategy. In addition, the typical microstructure of the Fe—36Ni alloy/glass-ceramic ultrafast laser micro-welds was characterized. The microstructural evolution and strengthen mechanism of as-produced whisker were systematically investigated. This study also provides an in-depth understanding into the co-ordination mechanisms of whiskers at the metal/glass-ceramic interface, which will broaden the scope of applications for composite components.

2. Experimental procedure

Ultrafast laser micro-welding technology was used in the joining of Glass-ceramic and Fe—36Ni alloy in an overlap configuration. Glass-ceramic Zerodur® with the dimension of 10mm×10mm×3mm and Fe—36Ni alloy with the dimension of 20mm×15mm×3mm was used. The chemical compositions of these materials were presented in [Table 1](#). Zerodur® is a typical representative of the LAS system of β -quartz solid solution glass-ceramics [1] with the coefficient of thermal expansion and melting point at zero pressure of 0.024ppm/k and 1570°C, respectively; while, the coefficient of thermal expansion and melting point of Fe—36Ni alloy are 1.99ppm/k and 1430°C, respectively ([Table 2](#)) [2,6].

Table 1. Chemical composition of Fe—36Ni and ZERODUR® Glass-ceramic.

Materials	Element (at.%)						
	C	Si	Mn	P	Ni	Fe	others
Fe-36Ni	0.01	0.12	0.42	0.006	35.32	64.12	0.004
ZERODUR®	Mg	P	Zr	Al	Si	O	others

Materials	Element (at.%)						
	C	Si	Mn	P	Ni	Fe	others
	0.79	1.38	2.23	11.39	21.13	61.34	1.74

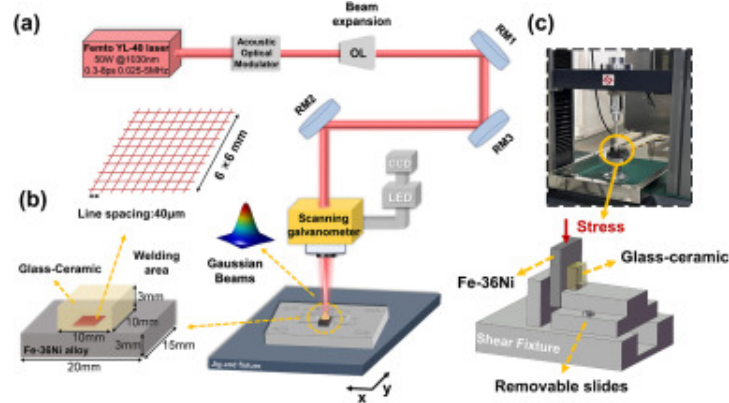
Table 2. Physical properties of Fe—36Ni and ZERODUR® Glass-ceramic [6,33].

Material properties	Fe-36Ni	ZERODUR® Glass-ceramic
Coefficient of thermal expansion (ppm/K)	1.99	0.024 (25°C)
Thermal conductivity λ (W/ (m.K))	12.56	1.46
Melting point (°C)	1430	966 (crystallization temperature T_c) 1570 (melting temperature T_m)
Density (g/cm ³)	8.03	2.53

Before welding, 400–2000 mesh diamond discs and a 2.5 μm polishing suspension were used for grinding and polishing the surface of Fe—36Ni alloy samples. Subsequently, the glass-ceramic and alloy components were ultrasonically cleaned in absolute ethanol for 10min to remove surface contaminants, including various particles and organic matter, before blowing dried.

Fig. 2 (a) shows a schematic diagram of the ultrafast laser micro-welding of glass-ceramic and Fe—36Ni alloy. The pulsed laser beam was initially directed into a beam expander via acoustic optical modulator, followed by several reflections through an aperture diaphragm to ensure peripheral stray light elimination. Finally, the laser beam was focused through the field lens. The laser beam with a Gaussian intensity profile was focused on the metal interface through glass-ceramic. In addition, a relatively large gap

between the two materials resulted in the escape and ablation of the plasma, rather than weld results. Therefore, the glass-ceramic was placed on the top of Fe—36Ni alloy in a fixture for flexible adjustment of the interface gap to achieve an intact tough between the materials, as sketched in Fig. 2 (b). More precisely, the gap was limited to within $1/4$ of the shortest visible wavelength, if the interface was observed to appear between the bright and dark concentric rings, that was, to achieve optical contact [34,35]. Then the samples were mounted using a fixture onto a computer-controlled x-y-z motion stage that allowed translation parallel or perpendicular to the laser propagation axis. With the z-direction adjusted manually using the z-axis. Ultrafast laser micro-welding was performed using the Femto YL-40 (Wuhan Anyang Laser Technology Co.). The laser wavelength was 1030nm, maximum laser output power was 52W, the pulse width ranged between 300fs and 8ps, the repetition frequency was 0.025–5MHz. The preliminary optimized parameters for the process were as follows: pulse duration ~800fs, repetition frequency~200kHz, scanning speed ~25mm/s, focusing spot ~40 μ m in diameter, and scanning line spacing ~40 μ m. The scanning was performed in a horizontal and then vertical grid by a galvanometer scanner.



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Fig. 2. Schematic diagram of (a) ultrafast laser micro-welding. (b) layout and dimension of base materials. (c) layout for the tensile-shear testing.

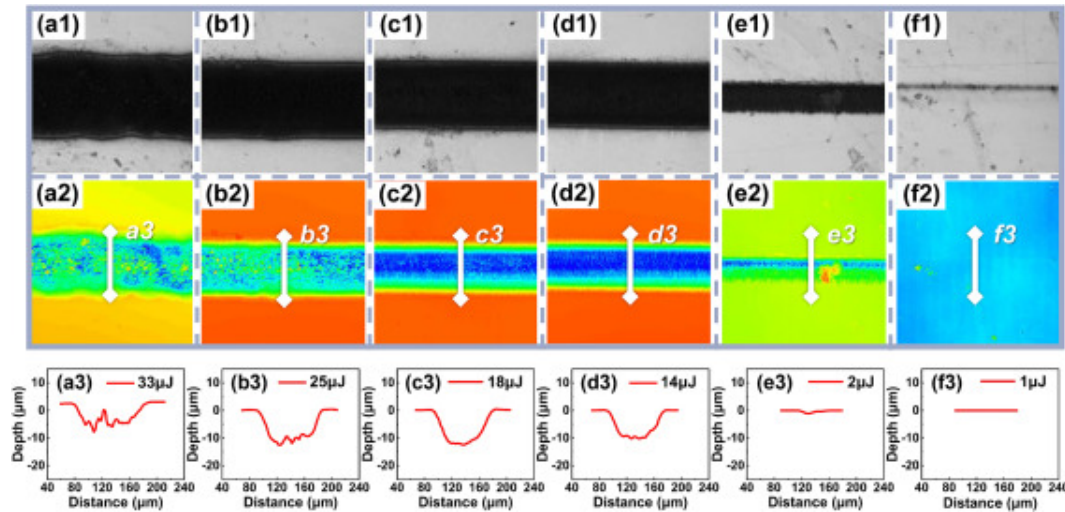
After ultrafast laser micro-welding, the macroscopic morphology and the three-dimensional morphology of the Fe-36Ni/glass-ceramic interface was conducted using Confocal laser scanning microscope (CLSM, OLS5000, Japan). The X-ray diffraction (XRD, D8 Advance, Germany) were utilized to determine the crystal structure of the samples before and after welding. The XRD analysis was accomplished at a scanning speed of $1^{\circ}/\text{min}$, Cu K α radiations produced at 40kV and 40mA with a 0.02° per second in the 2θ range of 20° to 90° . The microstructure of the cross sections was conducted using scanning electron microscope (SEM, ZEISS Gemini 300, Germany) equipped with energy dispersive spectroscopy (EDS, AMETEK). Mechanical performance of the joints was characterized by tensile-shear testings using a high-precision materials testing machine (Zwick, Zwick/Roell Z005, German) by applying a thrust force in a downward direction at a stable displacement rate of 0.5mm/min. Fig. 2(c) shows the schematic diagram of both the specimen and the fixture for tensile-shear testings. A minimum of three samples were tested for determining the average value across each experimental parameter. The surface morphology of fractures was conducted by SEM (ZEISS Sigma 300, Germany).

3. Results

3.1. Laser-induced ablation threshold evaluation

As known to all, after irradiated by ultrafast laser with a certain energy input, the material surface is normally ablated or modified to generate a mixture of gas, liquid, and plasma. Consequently, there is a threshold effect in laser ablation of materials. To validate the metal plasma sputtering during laser ablation and to anchor the adhesion behavior of the molten material under the influence of processing parameters, the laser-induced ablation threshold (LIAT) measurements were performed on the front surface of the Fe—36Ni alloy and the back surface of the glass-ceramic using the scribing method. Fig. 2, Fig. 3 show the morphological characteristics of ablation lines on the surface of Fe—36Ni alloy and glass-ceramic for different laser energies, respectively. The three-dimensional morphology and profile curve of the ablation lines were measured by means of an ultra-

depth of field microscope. Microgrooves were formed in the ablation regions, and it was found, as expected, that the width of microgrooves increased with increasing laser energy inputs.

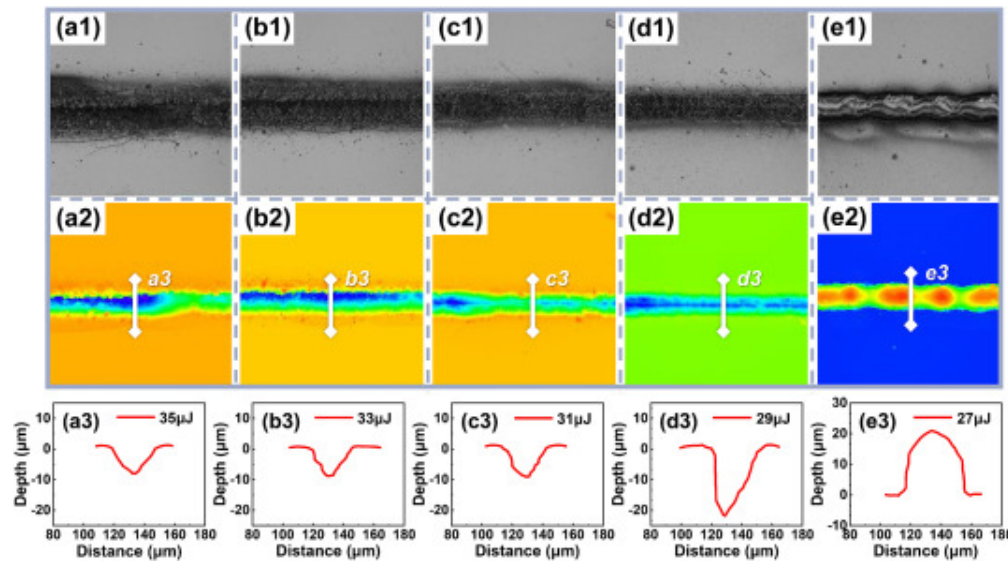


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Fig. 3. Laser-ablated morphologies on the front surface of Fe—36Ni at different pulse energies: (a1) 33μJ. (b1) 25μJ. (c1) 18μJ. (d1) 14μJ. (e1) 2μJ. (f1) 1μJ. (a2-f2) three-dimensional depth morphology of the ablation lines. (a3-f3) the profile curve of the microgroove.

With the laser energy decreased from 33μJ to 1μJ, the microgroove width of the Fe—36Ni alloy was decreased from 114μm to 11μm (Fig. 3), while the microgroove width of the glass-ceramic was significantly reduced from 51μm to 44μm when the laser energy decreased from 35μJ to 27μJ (Fig. 4). Notably, a distinct protrusion was observed on the back surface of the glass-ceramic when the laser energy was set at 27μJ (Fig. 4 (e1)), which was likely attributed to the incomplete penetration of the transparent material, resulting in the cooling and solidification of partially molten glass-ceramic after laser irradiation.



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Fig. 4. Laser-ablated morphologies on the back surface of glass-ceramic at different pulse energies: (a1) 35 μJ. (b1) 33 μJ. (c1) 31 μJ. (d1) 29 μJ. (e1) 27 μJ. (a2-e2) three-dimensional depth morphologies of the ablation lines. (a3-e3) the profile curve of the microgroove.

The expected $1/e^2$ focused spot size ($2\omega_0$) at the substrate surface can be calculated from the following Eq. (1) [36]:

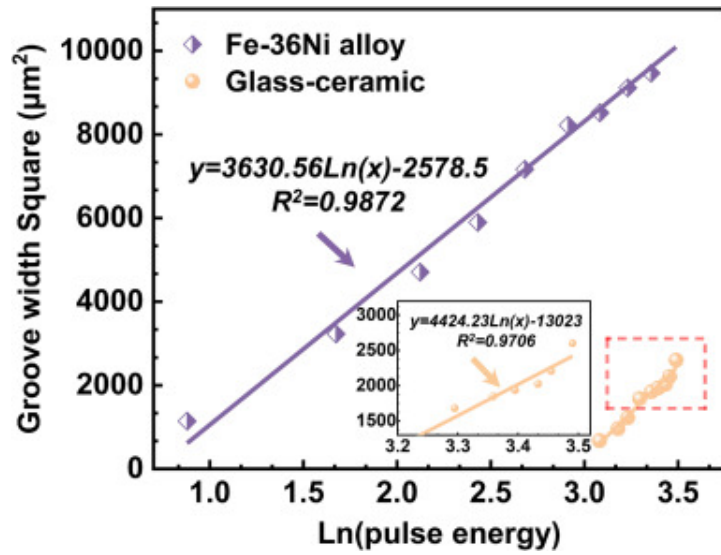
$$2\omega_0 = \frac{4}{\pi} \cdot \lambda \cdot \frac{f}{d} \cdot M^2 \quad (1)$$

where the wavelength (λ) is 1030 nm, the focal length of the f-theta lens (f) is 100 mm, the diameter of expanded laser beam (d) is 8 mm and the beam quality factor (M^2) is 1.3. Using these values, the focused waist diameter $2\omega_0 = 21.3 \mu\text{m}$.

Since the beam is near Gaussian distribution, the microgroove width D is related to the irradiation laser energy E by Eq. (2): [37]

$$D^2 = 2 \cdot \omega_0^2 \cdot \ln(E/\Phi_{th}) \quad (2)$$

where Φ_{th} is the ablation threshold energy, the damage curves of each sample are presented in Fig. 5.



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Fig. 5. Plot of microgroove width squared versus laser energy density.

It was noted that the LIAT corresponds to the maximum laser energy density at which the extrapolated ablation probability is equal to zero [38]. It should be emphasized that the theoretical calculation of the ultrafast laser ablation thresholds is based on the assumption of the free-electron density without taking into account the material's phase-transition mechanism and other factors. Finally, the LIAT of the Fe—36Ni alloy was fitted to be 0.57J/cm², and the glass-ceramic was measured to be 5.32J/cm². The coefficient of determination (R^2) for the LIAT evaluation were 0.9872 and 0.9706, respectively.

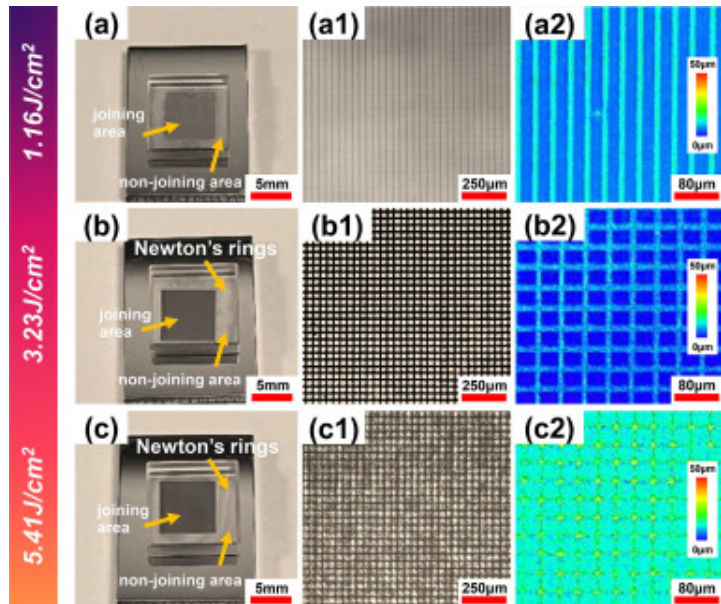
3.2. Macroscopic topography

Based on laser-induced material ablation thresholds, different laser energy density was deliberately selected, viz., $0.43\text{J}/\text{cm}^2$ (below the LIAT of Fe—36Ni alloy), $1.16\text{J}/\text{cm}^2$ and $3.23\text{J}/\text{cm}^2$ (above the LIAT of Fe—36Ni alloy but below the LIAT of glass-ceramic) and $5.41\text{J}/\text{cm}^2$ (above the LIAT of glass-ceramic), as shown in Table 3. The results showed that all the joints were successfully bonded with the exception of the LIAT of $0.43\text{J}/\text{cm}^2$. This was attributed to the absence of significant material modification, almost no splashing or adhesion of plasma was generated to fill the gap at the original interface. Therefore, the welds obtained at the LIAT of $1.16\text{J}/\text{cm}^2$, $3.23\text{J}/\text{cm}^2$ and $5.41\text{J}/\text{cm}^2$ were chosen for the further analysis.

Table 3. LIAT regulation strategy for the ultrafast micro-welding experiments.

Materials	Fe-36Ni			Glass-ceramic
LIAT (J/cm^2)	0.57			5.32
Laser energy density (J/cm^2)	0.43	1.16	3.23	5.41
Joining state	No	Yes	Yes	Yes

Fig. 6 exhibits the macroscopic morphologies of Fe—36Ni alloy/glass-ceramic joints under different laser energy density, at a repetition frequency of 200kHz, a pulse width of 800fs, and a line scan spacing of $40\mu\text{m}$. From the macrostructure as shown in Fig. 6, all the joints were well joined with absence of macro crack or deformation. Because of the ablation by femtosecond laser, the area of the laser-focused interface was darkened. It was interested to noted that the darkness of the joined area increased with the increase of laser energy density (Fig. 6 (a1-c1)). In addition, the interfacial region exhibited a tendency to fluctuate and undulate periodically with the increasing of the laser energy density from $1.16\text{J}/\text{cm}^2$ to $5.41\text{J}/\text{cm}^2$. Based on the corresponding three-dimensional morphology analysis, the heights at the laser-scanned paths were obviously greater than those in the untreated region (Fig. 6 (a2-c2)).



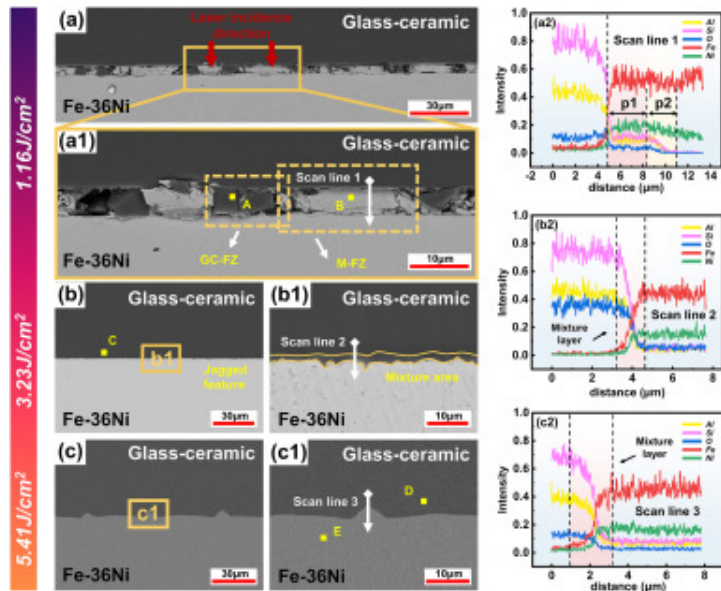
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Fig. 6. The macroscopic morphologies of Fe-36Ni/glass-ceramic joints at different pulse energy density: (a-a2) 1.16 J/cm^2 . (b-b2) 3.23 J/cm^2 . (c-c2) 5.41 J/cm^2 . (a2 -c2) are the three-dimensional morphology of the joined interface.

3.3. Interfacial microstructure

Fig. 7 shows typical microstructure of ultrafast laser micro-welds with different laser energy density. Since laser energy densities of 1.16 J/cm^2 (below glass-ceramic ablation threshold), 3.23 J/cm^2 (below glass-ceramic ablation threshold) and 5.41 J/cm^2 (above glass-ceramic ablation threshold) were used, which exhibited distinct interfacial characteristics. Further analysis at the interfacial region was performed to clarify the microstructure and composition of the micro-welds.



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Fig. 7. (a)~(c) The typical microstructure of Fe-36Ni/glass-ceramic ultrafast laser micro-welds.

When the laser energy density (1.16 J/cm^2) was between the ablation thresholds of glass-ceramics and Fe—36Ni alloys, only the metal substrate interacted with the laser, and no obvious modification region was found on the glass-ceramic side (Fig. 7 (a)). The direction of laser irradiation was shown by the arrow in Fig. 7 (a), the micro-welds domain would be divided into two types including grey area and black area. Fig. 7 (a1) shows the magnified image of area in Fig. 7 (a), combined with the EDS results in Table 4. According to the chemical composition analysis, it was inferred that metal fusion zone (M-FZ) and glass-ceramic fusion zone (GC-FZ) were generated. The M-FZ represents the recast layer of Fe—36Ni alloy, while the GC-FZ appears to be the re-solidification of molten glass-ceramic. In M-FZ, two regions of significant elemental distribution, named P1 and P2, were identified from EDS line scanning 1 (Fig. 7 (a2)). A small amount of glass-ceramics

was presented in the P1 region and remained at a stable level ($3.2\mu\text{m}$), this was due to the fact that the metal plasma was ejected outward from the surface by the intense interaction and recoil pressure caused by the laser movement. The expanded high-temperature plasma adhered to the back surface of the confining glass-ceramic. The generated heat was rapidly diffused to interact with the glass-ceramic, and the modified glass-ceramic containing metal nanoparticles acted as new laser energy absorption center, which splashed outward from the laser focus to both sides by surface tension [33]. At the same time, owing to the constraints of the back surface of the glass ceramic, the deposited nanoparticles continued to stack upwards and filled the original gap, the closer to the metal substrate, the less glass-ceramic content in the P2 region, eventually decreased to 0. In addition, some incompletely filled zones were clearly seen (Fig. 7 (a)), and these unfilled areas might be attributed to the poor fluidity of glass-ceramic at low laser energy density. When the laser energy reached $3.23\text{J}/\text{cm}^2$, the interface changed into a jagged feature without obvious micro voids or microcracks (Fig. 7 (b)). In addition, a mixture area (light grey region) was clearly seen at the interface (Fig. 7 (b1)). Based on the EDS line scanning 2 at the interfacial area, an element-diffusion-layer with the width of $1.4\mu\text{m}$ was observed in the pink region in Fig. 7 (b2).

Table 4. EDS compositions within the regions marked in Fig. 7 (A-E) in at. %.

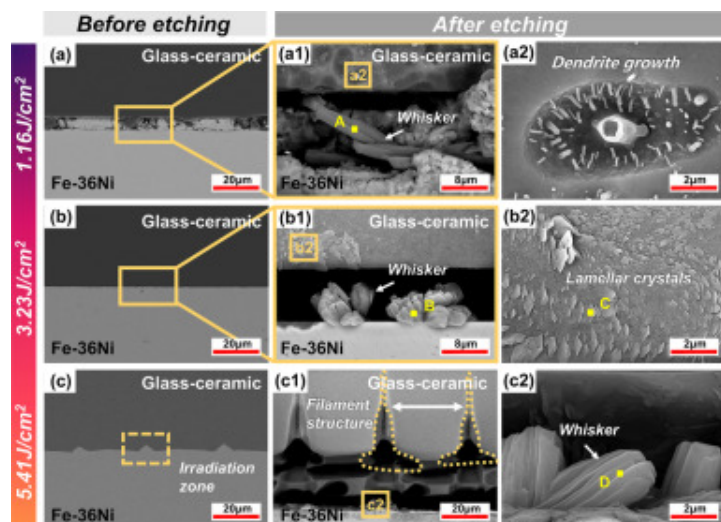
Regions	Elements				
	Fe	Ni	Al	Si	O
A	1.02	0.23	0.33	92.52	5.90
B	60.41	32.46	0.63	2.59	3.91
C	15.13	7.09	13.06	21.91	42.81
D	3.53	1.38	19.04	38.93	37.12
E	62.93	34.05	1.67	0.00	1.35

When the laser energy density reached 5.41 J/cm^2 , a regular volcano-like structure was formed in Fig. 7 (c) where Fe—36Ni alloy penetrated into the glass-ceramic, resulting in a mechanical interlocking at the interface (Fig. 7 (c1)). The EDS line scanning 3 revealed enhanced elemental diffusion at the interface, and the mixture region (pink region) was about $2.2 \mu\text{m}$ in width (Fig. 7 (c2)).

3.4. Whisker characterization

To investigate the diffusion-dominated interfacial evolution under varying laser energy densities, phase-selective etching was employed to isolate stable crystalline from the amorphous matrix, which enabled the characterization of crystalline embedded within the glass matrix. In this context, the residual glass phase, which serves as the continuous matrix in the glass-ceramic composites, was chemically targeted. Hydrofluoric acid (HF) is an effective candidate for dissolving the amorphous phase characterized by silicon-oxygen tetrahedra [39]. To resolve the distribution of whiskers within the micro-welds, the amorphous phase was etched using 40% w/w HF (AR grade), followed by rinsing for 5 min and drying with oil-free air for 30s.

Fig. 8 shows the SEM images of interfacial region of the ultrafast laser micro-welds at various laser energy density before and after etching. Fig. 8 (a) displayed some unfilled zones or defects at a laser energy density of 1.16 J/cm^2 . The precipitated crystal whiskers presented a needle-like structure after etching (Fig. 8 (a1)). In addition, dark grey oval craters were observed on the glass-ceramic near the interface in Fig. 8 (a1). Fig. 8 (a2) (the magnified image of area in Fig. 8 (a1)) illustrates a large number of needle-like structures, which were stripped from the parent material and distributed around the center white columnar crystals.



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Fig. 8. Cross-sectional SEM surface morphologies before and after etching with different laser energy density: (a-a2) 1.16J/cm². (b-b2) 3.23J/cm². (c-c2) 5.41J/cm².

As depicted in Fig. 8 (b), an increase in laser energy density to 3.23J/cm² resulted in micro-welds with robust mixing and diffusion. After etching, whiskers with a width of 4μm were appeared at the interface (Fig. 8 (b1)), and the whiskers grew towards the interface in different directions. Additionally, the glass-ceramic matrix produced a randomly orientated crystal structure of interlocking lamellar morphology in Fig. 8 (b2), suggesting an enhancement effect on the fracture toughness of glass-ceramic [40].

When the laser energy density was further increased to 5.41J/cm², regular mechanical interlocking structures were formed with no voids at the interface (Fig. 8 (c)). Regular filamentary structures were observed in Fig. 8 (c1). It is reported that these homogeneous filamentary structures will effectively improve the bonding strength [41]. Similarly, Fig. 8 (c2) exhibits a tightly bonded long-column structure at the interface. The EDS analysis of

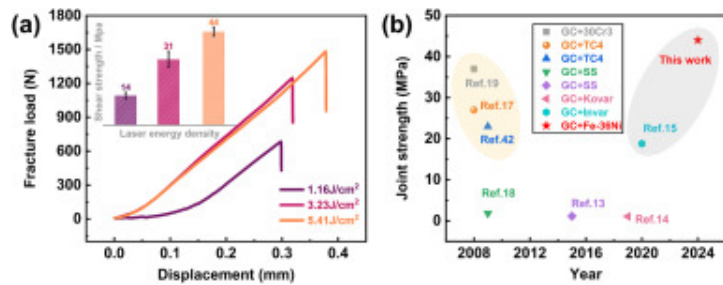
these whiskers reveals an elemental composition of 14.63 Fe-8.95 Ni-10.41 Al-15.61 Si-50.40 O (Table 5).

Table 5. EDS compositions within the regions marked in Fig. 8 (A-D) in at. %.

Regions	Elements				
	Fe	Ni	Al	Si	O
A	14.78	15.76	3.79	12.06	53.61
B	18.38	3.19	10.53	14.17	53.73
C	1.89	1.32	13.42	23.27	60.10
D	14.63	8.95	10.41	15.61	50.40

3.5. Mechanical property

The shear testings were performed to investigate the effect of laser energy density based on the LIAT on the mechanical properties of the ultrafast laser micro-welds. The selected laser energy densities were 1.16J/cm², 3.23J/cm² and 5.41J/cm², respectively, and the resulting of the ultrafast laser micro-welds shear strengths were shown in Fig. 9. The shear strength increased from 14MPa to 31MPa as the laser energy density increased from 1.16J/cm² to 3.23J/cm², and then further increased to 44MPa when the laser energy density was above the LIAT of glass-ceramic. The enhanced joint strength was compared with representative reports characterizing the shear strength of glass ceramic/metal [[13], [14], [15],[17], [18], [19],42], as seen in Fig. 9 (b). The maximum strength of the ultrafast laser micro-welds was the highest among all the currently published reports.



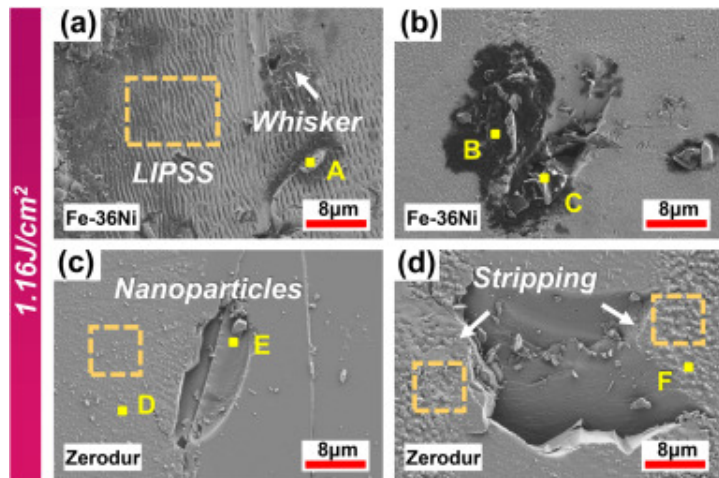
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Fig. 9. (a) Load-displacement of ultrafast laser micro-welds with various laser energy density. (b) the joint strength of Fe—36Ni alloy/glass-ceramic ultrafast laser micro-welds versus prior representative reports. The inset in (a) displays the shear strength of the ultrafast laser micro-welds.

3.6. Fractography

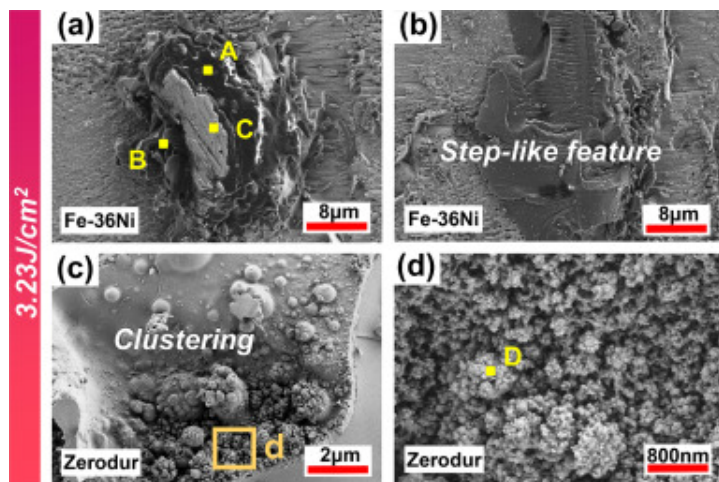
To investigate the representative fracture surface morphology of the ultrafast laser micro-welds, Fe—36Ni alloy/glass-ceramic micro-welds at different laser energy densities (1.16 J/cm², 3.23 J/cm², 5.41 J/cm²) were analyzed as shown in [Fig. 10](#), [Fig. 11](#), [Fig. 12](#).



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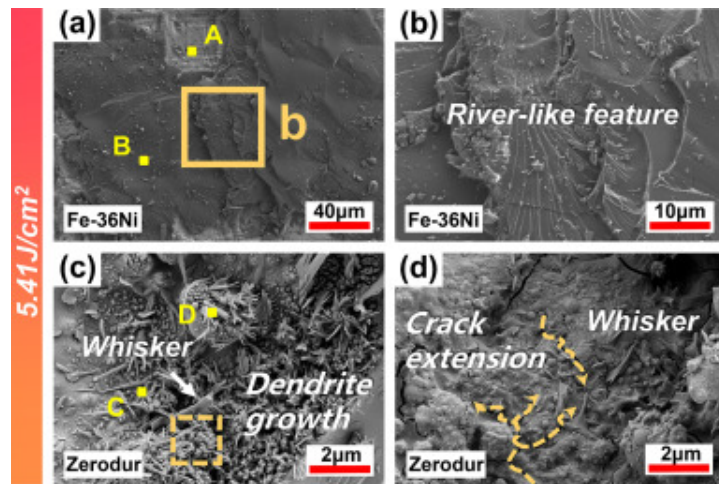
Fig. 10. SEM analysis on the fractured surfaces of the ultrafast laser micro-welds with 1.16 J/cm^2 laser energy density: (a-b) the surfaces of the Fe-36Ni separation. (c-d) the surfaces of the glass-ceramic separation.



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Fig. 11. SEM analysis on the fractured surfaces of the ultrafast laser micro-welds with $3.23\text{J}/\text{cm}^2$ laser energy density: (a-b) the surfaces of the Fe—36Ni separation. (c-d) the surfaces of the glass-ceramic separation.



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Fig. 12. SEM analysis on the fractured surfaces of the ultrafast laser micro-welds with $5.41\text{J}/\text{cm}^2$ laser energy density: (a-b) the surfaces of the Fe—36Ni separation. (c-d) the surfaces of the glass-ceramic separation.

Fig. 10 shows the representative fracture surface of ultrafast laser micro-welds at $1.16\text{J}/\text{cm}^2$. Fig. 10 (a-b) illustrate the fracture surface at Fe—36Ni alloy side. In Fig. 10 (a), a large area of ripple-like structure was observed. That was because the pulse energy density $1.16\text{J}/\text{cm}^2$ was above the LIAT of Fe—36Ni alloy which was high enough to create significant ablation and laser-induced periodic surface structures (LIPSS) [43]. The EDS results at point A suggested that the Fe—36Ni alloy surface contained a slight amount of Al and Si residuals, which was mainly originated from glass-ceramic (Table 6).

Furthermore, black and white areas were observed on the surface of the Fe—36Ni alloy

from the Fig. 10 (b), which was identified to be the mixing of Fe—36Ni alloy and glass-ceramic, respectively, by the EDS analysis (Table 6).

Table 6. EDS compositions within the regions marked in Fig. 10 (A-F) in at. %.

Regions	Elements				
	Fe	Ni	Al	Si	O
A	54.80	32.31	0.42	1.13	11.34
B	73.99	13.62	1.84	2.46	8.09
C	0.73	0.26	10.35	19.29	69.37
D	0.24	0.09	19.01	38.72	41.94
E	1.23	0.21	15.80	33.41	49.35
F	4.06	0.45	14.39	28.57	52.53

Fig. 10 (c-d) illustrate the fracture surface at glass-ceramic side. Under this condition ($1.16\text{J}/\text{cm}^2$), even if the laser energy density was sufficiently high to induce significant ablation on the Fe—36Ni alloy surface, the alloy particles adhering to the glass-ceramic side were unlikely to cause melting of glass-ceramic. Therefore, the interfacial fracture surface was observed to be relatively flat with no obvious dragged feature in Fig. 10 (c), and a small amount of nano-sized particles was attached to the glass-ceramic substrate, surrounded by partially molten glass-ceramic regions as supported by the EDS analysis (Table 6). In addition, small pieces of glass-ceramic were found to be peeled off, which might be attributed to the thermal melting. Additionally, incompletely separated nanoscale embryos were detected on the both sides of the peeled surface, as shown in Fig. 10 (d).

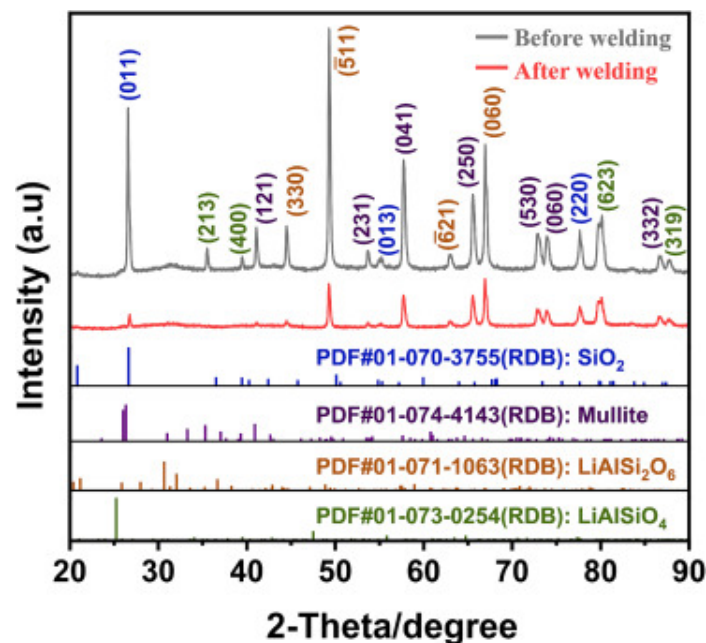
Fig. 11 shows the representative fracture surface morphology of ultrafast laser micro-welds at $3.23\text{J}/\text{cm}^2$. The fracture surface at Fe—36Ni alloy side were illustrated in Fig. 11

(a-b). Initially, Fig. 11 (a) shows a number of molten regions on the metal surface, accompanied by the accumulation of metallic debris, and the presence of abundant glass-ceramic adhesion on the metal side was confirmed according to the EDS results (Table 7). Furthermore, in Fig. 11 (b), the fracture surface was predominantly characterized by discernible cleavage steps, with a minor fraction of solidified glass-ceramic, reflecting the hard-brittle characteristic of the glass-ceramic.

Table 7. EDS compositions within the regions marked in Fig. 11 (A-D) in at. %.

Regions	Elements				
	Fe	Ni	Al	Si	O
A	1.04	0.35	10.16	18.76	69.69
B	21.57	7.43	15.75	29.52	25.73
C	59.05	34.15	0.70	0.66	5.44
D	1.12	0.38	13.27	28.05	57.18

The fracture surface within the glass-ceramic substrate is shown in Fig. 11 (c-d). In Fig. 11 (c), certain stripped nanocrystalline were presented a phenomenon of local aggregation, with some particles increased to about sizes of 2 μ m, and smaller particles (100–200nm in size) were distributed among the larger aggregates as magnified image in Fig. 11 (d). According to the EDS analysis, the atomic ratios of Al, Si, and O elements at point D were measured to be close to 1:2:6, which was tentatively suggested to be the LiAlSi₂O₆ phase. However, it must be emphasized that the lithium element cannot be accurately detected by EDS. When combined with the XRD pattern from the glass-ceramic side (Fig. 13), this phase was conclusively identified as the LiAlSi₂O₆.



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Fig. 13. XRD patterns of the glass-ceramic side before and after welding.

Fig. 12 shows the representative fracture surface morphology of ultrafast laser micro-welds at 5.41 J/cm^2 . Fig. 12 (a-b) illustrate the fracture surface at Fe—36Ni alloy side. Specifically, in Fig. 12 (a), residual glass-ceramic was observed to stand exactly upon the scanning tracks, along with river-like feature indicative of brittle fracture, as evidenced by the tear ridges in Fig. 12 (b).

The fracture surface at the glass-ceramic side is illustrated in Fig. 12 (c-d). In contrast to the structure in Fig. 11 (c), it was seen in Fig. 12 (c) that extremely fine dendritic embryos were in-situ formed on the glass-ceramic substrate at various angles except that of the distribution of nanocrystalline in the glass-ceramic matrix was relatively dispersive and uniform. The EDS analysis (Points C and D in Table 8) revealed that $\text{LiAlSi}_2\text{O}_6$ whiskers were evolved from the nano-sized particles. This observation was in agreement with the

report by Wang et al. [44,45]. Specifically, the primary axis of these dendritic branches typically extends to a length of approximately $2\mu\text{m}$ [46], with the grown $\text{LiAlSi}_2\text{O}_6$ whiskers displaying one-dimensional needle-like morphologies. As indicated by the arrows in Fig. 12 (c), these interwoven whiskers were demonstrated to reinforce joint bonding, as evidenced by their characteristic fractured cross-sections [47].

Table 8. EDS compositions within the regions marked in Fig. 12 (A-D) in at. %.

Regions	Elements				
	Fe	Ni	Al	Si	O
A	22.04	11.56	10.73	16.88	38.79
B	0.46	0.12	11.50	22.33	65.59
C	0.25	0.00	16.19	43.09	40.47
D	0.57	0.00	21.43	55.04	22.96

Furthermore, microcracks were identified in Fig. 12 (d), and a cross-linking interlock structure was found to distribute within crack extension zones, which might be more effective in blocking crack propagation. When the crack propagated through the $\text{LiAlSi}_2\text{O}_6$ phase, it encountered resistance from the phase itself [48,49]. Considering the laser processing, it should be noted that elevated energy density ($5.41\text{J}/\text{cm}^2$) enables sufficient thermal input for crystal growth [50]. Such microstructural evolution aligns with reported mechanisms where needle-like whiskers enhance fracture toughness and flexural strength through bridging effects [51,52]. Consequently, the improved shear strength of micro-welds was achieved as high as $44\pm 2\text{MPa}$ in Fig. 9.

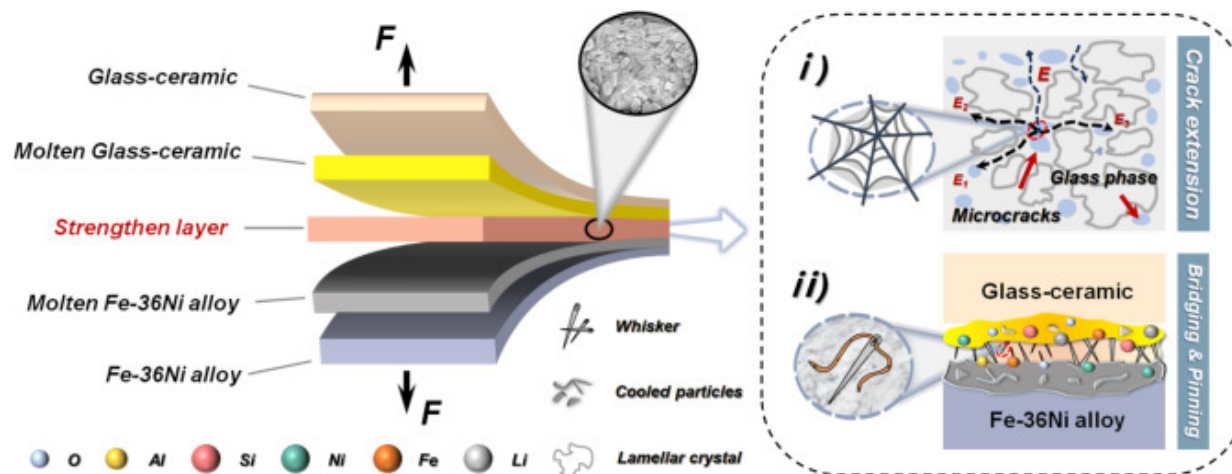
To further characterize the structural evolution of the phases within the weld, X-ray diffraction patterns were systematically performed on fracture specimens (Fig. 13). The primary crystalline phase compositions before (grey curve) and after (red curve) welding

were quantitatively determined by fitting the full XRD spectrum using Jade 6.0 software. As shown, four distinct crystalline phases: fused silica (SiO_2 , PDF#01-070-3755), mullite (Al_2SiO_5 , PDF#01-074-4143), eucryptite (LiAlSiO_4 , PDF#01-073-0254), and spodumene ($\text{LiAlSi}_2\text{O}_6$, PDF#01-071-1063) were determined, validating the individual phases in the micro-welds. Notably, the intensity of the diffraction peak was quite weak after welding, indicating the drop in crystal content. It was considered that the partly crystal phase experienced a melt quenched process and transformed into the corresponding amorphous phase. Furthermore, peak broadening and slight shifts were exhibited in the post-welding spectrum for some phases (e.g., (060) of eucryptite), indicating that lattice strain and crystallite size reduction were induced after welding.

4. Discussions

4.1. Whisker strengthen mechanism

An ultrastrong Fe—36Ni alloy/glass-ceramic ultrafast laser micro-welds is obtained with the maximum shear strength up to 44MPa. The primary reasons for their high strength are primarily attributed to phase transformation-induced crack deflection and bridging and pinning by needle-like whiskers. [Fig. 14](#) illustrates the whisker-strengthen mechanism in $\text{Li}_2\text{O}-\text{Al}_2\text{O}_3-\text{SiO}_2$ glass-ceramics, and following strengthen mechanism are discussed:



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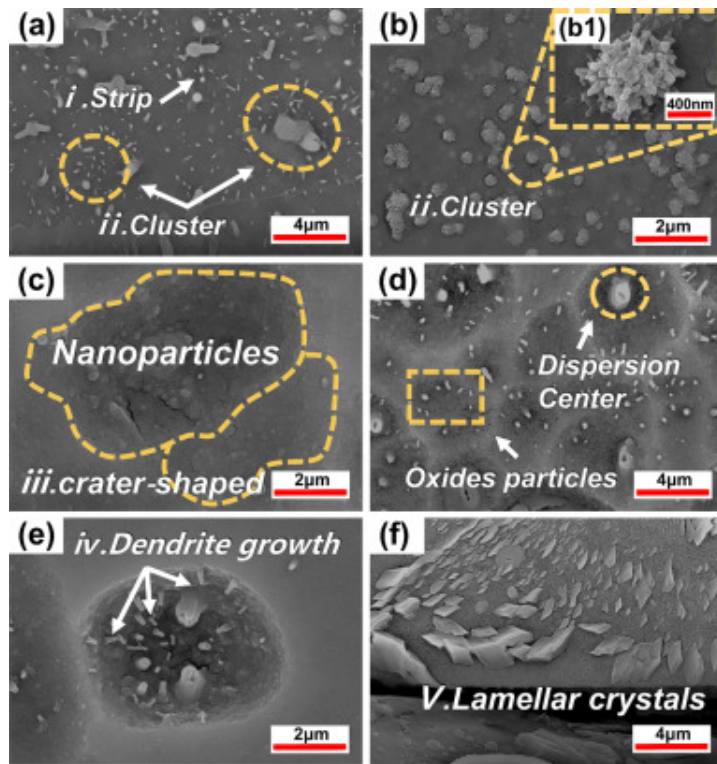
Fig. 14. The strengthen mechanism of the $\text{Li}_2\text{O}-\text{Al}_2\text{O}_3-\text{SiO}_2$ glass ceramics.

For one thing, due to the transformation of the β -quartz solid solution to the spodumene phase, the differential in the coefficient of thermal expansion frequently leads to the interfacial region between grains becoming the initiation point for crack propagation [53]. Whereas, the presence of the diffuse distribution of microcracks within the matrix (Fig. 12 (d)), results in the deflection, bifurcation, or termination of the propagation paths of primary cracks [54], thereby augmenting both the length of the crack propagation routes and the energy required for their progression (Fig. 14 i)).

For another, with the increase in laser energy density, grain growth, and morphological transformation are induced, leading to the formation of $\text{LiAlSi}_2\text{O}_6$ whiskers, which grew from the interfaces between glass-ceramic and glass phase to the adjacent metal substrate (Fig. 8 (a1-c1)). As demonstrated by the arrow-indicated fracture surfaces in Fig. 12 (c), these interlocking whiskers are involved in “bridging” and “pinning” during crack propagation (Fig. 14 ii)), resulting in an increase in crack propagation resistance between the two fracture surfaces [53].

4.2. Whisker evolution mechanism

Fig. 15 reveals the microstructural transformation features on the surface of $\text{Li}_2\text{O-Al}_2\text{O}_3\text{-SiO}_2$ glass-ceramics after laser irradiation: i) stripping of oxide particles, ii) clustering, iii) collapse into crater-shaped, iv) dendrite growth and v) lamellar crystals from whiskers. Evolutionary processes of the whiskers are discussed as follows:



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Fig. 15. The microstructural evolution of the $\text{Li}_2\text{O-Al}_2\text{O}_3\text{-SiO}_2$ glass ceramics.

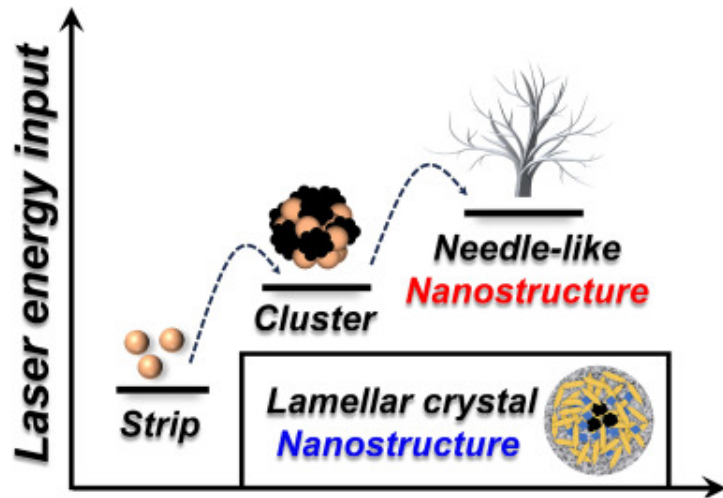
First, initial stage: multiple dispersed oxide particles on the glass-ceramic substrate was observed (Fig. 15 (a)), where these small particles is likely to be the starting point for whisker growth. The stripped nanoparticles are formed into small aggregates with

localized clusters in the early stage of their growth (Fig. 15 (b)), providing a foundation for subsequent whisker growth.

Second, grown stage: the local structure of the substrate is weakened by the precipitation of oxide particles, and these aggregated regions collapse towards the interior of the glass-ceramic, which is presented in Fig. 15 (c). Furthermore, the edges of these collapsed regions are surrounded by relatively larger particles in Fig. 15 (d), with a dispersion center at the crater formed. Fig. 15 (e) shows the dendritic growth process of whiskers; the crystals are transformed from spherical grains to columns of various sizes. The phenomenon of interfacial dendrite growth centered on oxide particles suggests that dispersed oxide particles serve as growing embryos for dendrite formation [45]. It is of notable significance that the presence of stress concentration or defects in the material results in a relatively high density of cracks within the crater region. The precipitated reinforcing phases (e.g. whiskers) exhibit a capacity to enhance the material's toughness by bridging across the crack interfaces and hindering their propagation.

Finally, mature stage: the lamellar crystal structure of the whisker reaches maturity where the crystal morphology undergoes a significant transformation (Fig. 15 (f)). This kind of morphology is likely to be attributed to the preferred orientation of lamellar, therefore the residual stress within the material is effectively reduced leading to the improvement on both the strength and the toughness.

Fig. 16 concludes the whisker evolution mechanism of $\text{Li}_2\text{O}-\text{Al}_2\text{O}_3-\text{SiO}_2$ glass ceramic: the process begins with the stripping of oxide particles, which subsequently aggregate into clusters and undergo a transformation, ultimately resulting in the formation of spherical crystals, and then, at higher temperatures, the whisker spodumene crystals grow at the cost of the small spherical crystals.



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Fig. 16. The evolution mechanism of the $\text{Li}_2\text{O}-\text{Al}_2\text{O}_3-\text{SiO}_2$ glass ceramic.

5. Conclusion

By using LIAT strategy, ultrastrong glass-ceramic and Fe—36Ni alloy joints by ultrafast laser micro-welding technology was obtained for the first time. The microstructure, mechanical properties, whisker strengthen and evolution mechanism of the ultrafast laser micro-welds were investigated. The primary conclusions were summarized as follows:

- (1) The characteristic damage curves are plotted and ablation thresholds for laser scanning processes were determined at 1030nm in wavelength, 800fs in pulse duration and 200kHz pulse repetition rate. The ablation thresholds for Fe—36Ni alloy and glass-ceramic were thus calculated to be $0.57\text{J}/\text{cm}^2$ and $5.32\text{J}/\text{cm}^2$ at a scanning speed of 25 mm/s, respectively.

- (2) Needle-like whiskers with high aspect ratios were precipitated at the micro-welds, and the XRD and EDS results were proved that these whiskers were $\text{LiAlSi}_2\text{O}_6$ phase. These whiskers interlaced each other to form a mechanical interlock structure, which was beneficial for the enhancement of the micro-welds with the maximum strength of 44MPa.
- (3) Joint strengthening is achieved by the following dual mechanisms: i) β -quartz-to-spodumene phase transformation generates interfacial microcracks for crack deflection, ii) $\text{LiAlSi}_2\text{O}_6$ whiskers provide crack pinning via interfacial bridging. These synergistic effects collectively improve fracture resistance and mechanical performance in the glass-ceramic composite system.
- (4) Laser-induced whisker evolution mechanism in $\text{Li}_2\text{O}-\text{Al}_2\text{O}_3-\text{SiO}_2$ glass-ceramics is divided into three primary stages: stripping of oxide particles, clustered into spherical crystalline aggregates, and progressively transformed into whisker-like morphologies under elevated temperatures.

CRediT authorship contribution statement

Tao Zhang: Writing – original draft, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Jin Yang:** Writing – review & editing, Validation, Resources, Funding acquisition. **Meng Yang:** Software, Methodology. **Ming Wu:** Data curation. **Rui Zhou:** Conceptualization. **Jianian Tian:** Data curation. **Yixuan Zhao:** Methodology. **Caiwang Tan:** Validation. **Xiaoguo Song:** Visualization.

Declaration of competing interest

No conflict of interest exists in the submission of this manuscript, and the manuscript is approved by all authors for publication. I would like to declare on behalf of my co-authors that the work described was original research that has not been published previously and

is not under consideration for publication elsewhere, in whole or in part. All the authors have approved the manuscript that is enclosed.

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